Test of hypothesis

Population: The aggregation of all the possible values of a variable.

Sample: A sample is the representative part of the population.

Random sample: A sample drawn in such a way that every individual in the population has same chance of being included in the sample.

Parameter: The unknown constants or any function of them that appears in the mathematical specification of a population are known -- " describe a characteristics of an entire population) (mean of total population)

statistic: Any function of a set of sample value.

Hypothesis: Any statement, description, specification about a population is called hypothesis.

statistical hypothesis: Any statement about the probability distribution of a set of a random variable.

- Null hypothesis: The statistical hypothesis picked up for testing. often written as $H_0 : \gamma = \gamma_0$.
- Alternative hypothesis: Any hypothesis other than H_0. written as $H_1 : \gamma \neq$

Test of hypothesis:

The statistical procedure used to verify any pre-specified statement or description regarding a population or population parameter.

Level of significance:

The probability of type-I error, denoted by α. It is the risk of rejecting H_0 even though it is true. Choosen by researcher before the test, usually 5% or 1%

Type of error:

a) Type I error: Rejecting H_0, when it is true. (α), convicting innocent people.

b) Type II error: Accepting H_0, when it is false. (β), Freeing a gilty person.

P value / Observed level of significance: The actual risk of commiting type-I error. if $p < \alpha \rightarrow$ test is significant, reject H_0

$p > \alpha \rightarrow$ do not reject H_0

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Important steps in a test of significance:

1) Set up the null hypothesis, state an appropriate H_0 (and H_1)
2) Make assumption
3) Choose the level of significance (α)
4) Calculate a suitable test statistics.
5) Find the critical value. 6) Compare and decide.

Statistical test:

The important test of significance are classified broadly as:

a) Normal test b) t-test c) χ^2 test d) F-tests.

a) **Normal test:** → "how many standard errors away"

If U is a statistics, then

$$Z = \frac{U - E(U)}{SE(U)} \sim N(0,1)$$

| used for testing hypothesis about mean, proportions.

b) **t-test:**

Normal case is the special case of t-test when n (degree of freedom) is large.

So, t-test sometimes called a small-sample test.

$$t = \frac{U - EU}{\text{estimated } \sigma} \quad \text{with d.f. } n.$$

| used for means, regression coefficient

c) **χ^2 test:**

$$\chi^2 = \frac{\Sigma (O_i - E_i)^2}{E_i}$$

| O → observed frequency. E → Expected frequency

| frequency fit

d) **F-test:**

Let S_1^2 and S_2^2 be two independent estimates of the same variance σ^2 of a normal population, based on v_1 and v_2 degree of freedom.

$$F(v_1, v_2) = S_1^2 / S_2^2$$

| used to test the hypothesis.

Test of Significance about Mean

(a) population mean equal to specific value:

We have random sample of size n and test weather the population mean μ equals some assigned value μ_0.

Have 2 case arrive (i) σ known / large sample (ii) unknown / small sample.

Case(i) σ known, or estimated from a large sample ($n > 30$)

use the normal z test.

Hypothesis: • $H_0 \Rightarrow \mu = \mu_0$
• $H_1 \Rightarrow \mu \neq \mu_0$

assumption
1) Sample are randomly selected
2) The observation are drawn from a normal pop^n
3) σ are known / estimated from large sample

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

Critcal value (two tailed $\alpha = 0.05$): -1.96 and $+1.96$

Case(ii) σ unknown, and sample is small

here use the t-test. Hypothesis and assumption same as case (i)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad \text{with } (n-1) \text{ degree of freedom}$$

where, $s^2 = \frac{1}{n-1} \Sigma (x_i - \bar{x})^2$

Critial value $-t\left(n-1, \frac{\alpha}{2}\right)$ and $+t\left(n-1, \frac{\alpha}{2}\right)$

(ii) Comparison of two independent sample means:

Two independent random sample of sizes n_1 and n_2 from two populations. population are assumed normal with means μ_1, μ_2 and variance σ_1^2, σ_2^2. Sample mean and variance. $\bar{x}_1 = \frac{\Sigma x_{1i}}{n_i}$ • $s_1^2 = \frac{\Sigma (x_{1i} - \bar{x}_1)^2}{n_1 - 1}$

Hypothesis: $H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$

Case(I): Variance known, or samples large ($n_1 > 30$, $n_2 > 30$)

σ_1^2 and σ_2^2 are variance

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

if σ unknown the calculate s^2 Just use s^2 insted of σ^2, in case of large sample

Critical valve: two tailed (-1.96 and $+1.96$) which $\alpha = 0.05$

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Case (ii) : Small samples, variance unknown but assume equal

here the population variance are assumed equal.

$$t = \frac{\bar{m}_1 - \bar{m}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$, with (n_1+n_2-2) degree of freedom.

two tailed , $\alpha = 0.05$

where the pooled variance,

$$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1+n_2-2)}$$

Significance of correlation Coefficient

(i) Test of zero correlation:

hypothesis: • H_0: $\rho = 0$ (no linear correlation in the population)

• H_1: $\rho \neq 0$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$ | r = Correlation coefficient

with $(n-2)$ d.f.

Significance test in Regression Analysis ($y = \alpha + \beta x_i + e_i$)

i) Test of intercept (α)

Hypothesis: $H_0: \alpha = \alpha_0$; $H_1: \alpha \neq \alpha_0$ | $\alpha_0 = 0$

$$t = \frac{\alpha - \alpha_0}{\sqrt{\frac{\sigma^2 \times \Sigma x_i^2}{n\Sigma(x_i - \bar{x})^2}}}$$ with $(n-2)$ d.f.

ii) Test of slope (β)

hypothesis: $H_0: \beta = \beta_0$; $H_1: \beta \neq \beta_0$

$$t = \frac{\beta - \beta_0}{\sigma/\sqrt{\Sigma(x_i - \bar{x})^2}}$$ with $(n-2)$ df.

$$\sigma^2 = \frac{1}{n-2}\left[SS(y) - \beta\, SP(x,y)\right]$$

$$SS(y) = \Sigma y_i^2 - n\bar{y}_i^2$$

$$SP(x,y) = \Sigma x_i y_i - n\bar{x}\bar{y}$$

$$\Sigma(x_i - \bar{x}) = \Sigma x_i^2 - n\bar{x}^2$$